

Supplemental material

Oncogenicity classifications for 94 somatic variants

Gene symbol	Slide numbers
BRAF	2-11
EZH2	12-24
FLT3	25-34
IDH1	35-43
KRAS	44-47
PIK3CA	48-57
PTEN	58-67
RB1	68-78
TERT	79-85
TP53	86-95

BRAF p.Val600Lys (NM_004333.6:c.1798_1799delinsAA, hg19 chr7:g.140453136_140453137delinsTT)

- An SOP based classification: *Likely Oncogenic*
(2 Strong, 1 supporting) {9 points}
- Satisfied criteria
 - OS2, functional evidence PMID: 15035987; class 1 *BRAF* variant
 - OS3, AA 600 change count in cancerhotspots.org is ≥ 50 (897), p.Val600Lys count 24
 - OP4, absent from controls

Notes: Allele Registry ID: CA16602420; Relevant tumor type(s): neoplasms. (In PMID: 15035987 AA numbers shifted by one, AA 599 corresponds to AA 600.)

BRAF p.Lys601Glu (NM_004333.6: c.1801A>G, hg19 chr7:g.140453134T>C)

- An SOP based classification: *Likely Oncogenic*
(1 Strong, 1 moderate AND 2 supporting) {8 points}
- Satisfied criteria
 - OS2, functional evidence PMID: 15035987, 28783719; class 2 *BRAF* variant
 - OM3, AA 601 change count in cancerhotspots.org is ≥ 10 (38), p.Lys601Glu count 32
 - OP4, absent from controls {gnomAD v2.1.1 (non-cancer)}
 - OP1, FATHMM prediction pathogenic (score 0.98) and CADD prediction likely deleterious (score 24.5)

Notes: Allele Registry ID: CA123645; Relevant tumor type(s): neoplasms. (In PMID: 15035987 AA numbers shifted by one, AA 600 corresponds to AA 601.)

BRAF p.Gly469Ala (NM_004333.6: c.1406G>C, hg19 chr7:g.140481402C>G)

- An SOP based classification: *Likely Oncogenic*
(1 Strong, 1 moderate AND 2 supporting) {8 points}
- Satisfied criteria
 - OS2, functional evidence PMID: 15035987, 28783719; class 2 *BRAF* variant
 - OM3, AA 469 change count in cancerhotspots.org is ≥ 10 (61), p.Gly469Ala count 34
 - OP4, rare in controls {one entry in gnomAD v2.1.1}
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 27.5)

Notes: Allele Registry ID:CA123655; Relevant tumor type(s): neoplasms. (In PMID: 15035987 AA numbers shifted by one, AA 468 corresponds to AA 469.)

BRAF p.Gly469Glu (NM_004333.6: c.1406G>A, hg19 chr7:g.140481402C>T)

- An SOP based classification: *Likely Oncogenic*
(1 Strong AND 3 supporting) {7 points}
- Satisfied criteria
 - OS2, functional evidence PMID: 15035987, 18794803 (low kinase activity, signaling through CRAF), class 3 *BRAF* variant
 - OP4, absent in controls
 - OP3, p.Gly469Glu change count in cancerhotspots.org is < 10 (4)
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 33)

Notes: Allele Registry ID:CA279970; Relevant tumor type(s): neoplasms. (In PMID: 15035987 AA numbers shifted by one, AA 468 corresponds to AA 469.)

BRAF p.Asp594Asn (NM_004333.6:c.1780G>A, hg19 chr7:g.140453155C>T)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, functional evidence PMID: 28783719 (loss of kinase activity, signaling through CRAF), class 3 *BRAF* variant
 - OS3, AA 594 change count in cancerhotspots.org is ≥ 50 (51), p.Asp594Asn count 24
 - OP4, absent from controls
 - OP1, FATHMM prediction pathogenic (score 0.98) and CADD prediction deleterious (score 33)

Notes: Allele Registry ID: CA135095; Relevant tumor type(s): neoplasms.

BRAF p.Asn486_Pro490del (NM_004333.6:c.1457_1471del15, hg19 chr7:g.140477837_140477851del)

- An SOP based classification: *Oncogenic*
(1 Strong, 2 moderate AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, functional evidence PMID: 26732095
 - OM1, located in well-established functional domain (kinase domain)
 - OM2, protein length due to in-frame deletions/insertions in known oncogene
 - OP3, p.Asn486_Pro490del change count in cancerhotspots.org is < 10 (3)
 - OP4, absent in controls

Notes: Allele Registry ID:CA645556774; Relevant tumor type(s): neoplasms.

BRAF p.Pro403LeufsTer8 (NM_004333.6:c.1208del, hg19 chr7:g.140482927del)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, absent in controls

Notes: Allele Registry ID:CA458116210; Relevant tumor type(s): neoplasms.

BRAF p.Pro14Arg (NM_004333.6:c.41C>G, hg19 chr7:g.140624463G>C)

- An SOP based classification: *VUS*
(1 supporting {OP4} AND 1 supporting {BP1}) {0 points}
- Satisfied criteria
 - OP4, absent from controls
 - SBP1, FATHMM-XF prediction benign (score 0.066) and CADD prediction likely benign (score 11.81)

Notes: Allele Registry ID: CA281924; Relevant tumor type(s): neoplasms.

BRAF p.Ala31Gly (NM_004333.6:c.92C>G, hg19 chr7:g.140624412G>C)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - SBP1, FATHMM-XF prediction benign (score 0.106) and CADD prediction likely benign (score 14.21)

Notes: Allele Registry ID: CA135146; Relevant tumor type(s): neoplasms.

ClinGen RASopathy Variant Curation Expert Panel - Likely Benign

(<https://www.ncbi.nlm.nih.gov/clinvar/variation/44833/>)

BRAF p.Asp445His (NM_004333.6:c.1333G>C, hg19 chr7:g.140481475C>G)

- An SOP based classification: *VUS*
(2 supporting) {2 points}
- Satisfied criteria
 - OP4, absent from controls
 - OP1, FATHMM-XF prediction pathogenic (score 0.879) and CADD prediction deleterious(score 32)

Notes: Allele Registry ID: CA369589029; Relevant tumor type(s): neoplasms.

Zero entries in COSMIC. Outside of kinase domain.

EZH2 p.Tyr646Phe (NM_004456.4:c.1937A>T, hg19 chr7:g.148508727T>A)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, gain of function demonstrated by altered substrate specificity and increased catalytic efficiency for H3K27 trimethylation (PMID: 21190999, 24563539, 30692681)
 - OS3, hotspot in COSMIC and in cancerhotspots.org, AA position count in cancerhotspots.org is < 50 (38) and AA change count is ≥ 10 (13)
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 32)

Notes: Allele Registry ID: CA16602259; Relevant tumor type(s): neoplasms in tissues with *EZH2* expression

EZH2 (NM_004456.4:c.-7-2A>C, hg19 chr7:g.148544399T>G)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA916079855; Relevant tumor type(s): neoplasms in tissues with *EZH2* expression.

EZH2 p.Ala682Gly (NM_004456.4:c.2045C>G, hg19 chr7:g.148506467G>C)

- An SOP based classification: *Likely Oncogenic*
(1 Strong AND 1 moderate AND 1 supporting) {7 points}
- Satisfied criteria
 - OS2, p.Ala682Gly leads to increased H3K27 trimethylation in human B-cell lymphoma cell lines (PMID: 22323599) and in breast cancer cell lines (PMID: 25253781)
 - OM1, Located in a SET domain
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA16602672; Relevant tumor type(s): neoplasms in tissues with *EZH2* expression.

EZH2 p.Ala682Val (NM_004456.4:c.2045C>T, hg19 chr7:g.148506467G>A)

- An SOP based classification: *VUS*
(1 moderate AND 2 supporting) {4 points}
- Satisfied criteria
 - OM1, Located in a SET domain
 - OP4, Absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 32)

Notes: Allele Registry ID: CA369712350; Relevant tumor type(s): neoplasms in tissues with *EZH2* expression. P.Ala682Gly leads to increased H3K27 trimethylation in human B-cell lymphoma cell lines (PMID: 22323599) and in breast cancer cell lines (PMID: 25253781).

EZH2 p.Asp136Val (NM_004456.4:c.407A>T, hg19 chr7:g.148526897T>A)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA369721731; Relevant tumor type(s): neoplasms in tissues with *EZH2* expression. This is a residue that a number of structural studies point to be as being likely of functional importance.

EZH2 p.Asp233Gly (NM_004456.4:c.698A>G, hg19 chr7:g.148524286T>C)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA369719623; Relevant tumor type(s): neoplasms in tissues with *EZH2* expression. This variant has been reported in a patient with Weaver Syndrome (PMID 24214728), and this residue is PARylated by PARP1 (PMID 28925391).

EZH2 p.Arg658Thr (NM_004456.4:c.1973G>C, hg19 chr7:g.148507481C>G)

- An SOP based classification: *VUS*
(1 moderate AND 1 supporting) {3 points}
- Satisfied criteria
 - OM1, Located in a SET domain
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA369712693; Relevant tumor type(s): neoplasms in tissues with *EZH2* expression. Reduced methylation activity in assay with change to alanine at this AA position (PMID 29185984).

EZH2 p.Arg684Cys (NM_004456.4:c.2050C>T, hg19 chr7:g.148506462G>A)

- An SOP based classification: *Likely Oncogenic*
(1 Strong, 1 moderate AND 2 supporting) {8 points}
- Satisfied criteria
 - OS2, Impaired Histone Methyltransferase Function (PMID 26694085)
 - OM1, Located in a SET domain
 - OP4, Absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction pathogenic (score 0.97) and CADD prediction deleterious (score 35)

Notes: Allele Registry ID: CA271512; Relevant tumor type(s): myeloid malignancies

Has been observed to be de novo in individuals with clinical features of Weaver syndrome (PMID: 22190405, 26694085, 29802153).

EZH2 p.Arg690His (NM_004456.4:c.2069G>A, hg19 chr7:g.148506443C>T)

- An SOP based classification: *Likely Oncogenic*
(1 Strong, a moderate AND 2 supporting) {8 points}
- Satisfied criteria
 - OS2, loss of downstream H3K27 trimethylation activity (PMID: 20601953)
 - OM3, hotspot in COSMIC (not hotspot in cancerhotspots.org)
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 35)

Notes: Allele Registry ID: CA369712270; Relevant tumor type(s): myeloid malignancies

EZH2 p.Ser695Leu (NM_004456.4:c.2084C>T, hg19 chr7:g.148506428G>A)

- An SOP based classification: *VUS*
(1 moderate AND 1 supporting) {3 points}
- Satisfied criteria
 - OM1, Located in a SET domain
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA369712199; Relevant tumor type(s): neoplasms in tissues with *EZH2* expression. Has been identified in Weaver syndrome patients (PMID 24214728). Structural studies point to it as a substrate binding residue (PMID 24364637).

EZH2 p.Val626Met (NM_004456.4:c.1876G>A, hg19 chr7:g.148508788C>T)

- An SOP based classification: *Likely* Oncogenic (1 Strong AND 2 supporting) {6 points}
- Satisfied criteria
 - OS2, Impaired Histone Methyltransferase Function (PMID: 29244146)
 - OP4, Absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction pathogenic (score 0.98) and CADD prediction deleterious (score 34)

Notes: Allele Registry ID: CA271506; Relevant tumor type(s): myeloid malignancies. Has been observed to be de novo in individuals with clinical features of Weaver syndrome (SCV000588200.1)

EZH2 p.Tyr731Phe (NM_004456.4:c.2192A>T, hg19 chr7:g.148506166T>A)

- An SOP based classification: *Likely Oncogenic*
(1 Strong AND 1 moderate AND 1 supporting) {7 points}
- Satisfied criteria
 - OS2, Catalytically inactive (PMID 31488576)
 - OM1, Located in a SET domain
 - OP4, Absent from controls (gnomAD v2.1.1 non-cancer)

Notes: Allele Registry ID: CA369711887; Relevant tumor type(s): myeloid malignancies.

EZH2 p.Tyr731His (NM_004456.4:c.2191T>C, hg19 chr7:g.148506167A>G)

- An SOP based classification: *VUS*
(1 moderate AND 2 supporting) {4 points}
- Satisfied criteria
 - OM1, Located in a SET domain
 - OP4, Absent from controls (gnomAD v2.1.1 non-cancer)
 - OP1, FATHMM prediction pathogenic (score 1.00) and CADD prediction deleterious (score 31)

Notes: Allele Registry ID: CA369711890; Relevant tumor type(s): neoplasms in tissues with *EZH2* expression. Tyr731Phe variant is catalytically inactive (PMID 31488576).

FLT3 p.Asp835Tyr (NM_004119.3:c.2503G>T, hg19 chr13:g.28592642C>A)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, gain of function demonstrated by constitutive activation and transformation of cultured cells (PMID: 11290608, 15256420)
 - OS3, hotspot in COSMIC and in cancerhotspots.org, AA position count in cancerhotspots.org is < 50 (17) and AA change count is ≥ 10 (11)
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.99) and CADD prediction deleterious (score 34)

Notes: Allele Registry ID: CA126347; Relevant tumor type(s): hemopoietic neoplasms

FLT3 p.Asp835His (NM_004119.3:c.2503G>C, hg19 chr13:g.28592642C>G)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, gain of function demonstrated by constitutive activation and transformation of cultured cells (PMID: 11290608, 15256420)
 - OS3, hotspot in COSMIC and in cancerhotspots.org, AA position count in cancerhotspots.org is < 50 (17) and AA change count is < 10 (2)
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.99) and CADD prediction deleterious (score 33)

Notes: Allele Registry ID: CA280208; Relevant tumor type(s): hemopoietic neoplasms

FLT3 p.Asp835Val (NM_004119.3:c.2504A>T, hg19 chr13:g.28592641T>A)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, gain of function demonstrated by constitutive activation and transformation of cultured cells (PMID: 11290608, 15256420)
 - OS3, hotspot in COSMIC and in cancerhotspots.org, AA position count in cancerhotspots.org is < 50 (17) and AA change count is < 10 (1)
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.99) and CADD prediction deleterious (score 33)

Notes: Allele Registry ID: CA126344; Relevant tumor type(s): hemopoietic neoplasms

FLT3 p.Asp835Glu (NM_004119.3:c.2505T>A, hg19 chr13:g.28592640A>T)

- An SOP based classification: *Likely Oncogenic*
(1 Strong, 1 moderate AND 2 supporting) {8 points}
- Satisfied criteria
 - OS2, gain of function demonstrated by constitutive activation and transformation of cultured cells (PMID: 11290608, 15256420)
 - OM3, hotspot in COSMIC and in cancerhotspots.org, AA position count in cancerhotspots.org is < 50 (17) and AA change count is < 10 (3)
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.89) and CADD prediction deleterious (score 24.2)

Notes: Allele Registry ID: CA16602449; Relevant tumor type(s): hemopoietic neoplasms

FLT3 p.Asp835Ile (NM_004119.3:c.2503_2504delinsAT, hg19 chr13:g.28592641_28592642delinsAT)

- An SOP based classification: *VUS*
(1 moderate AND 1 supporting) {3 points}
- Satisfied criteria
 - OM1, located in well-established functional domain
 - OP4, absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA923726224; Relevant tumor type(s): hemopoietic neoplasms; potential drug resistance to *FLT3* inhibitors (PMID: 27908881)

FLT3 p.Asn676Lys (NM_004119.3:c.2028C>A, hg19 chr13:g.28602340G>T)

- An SOP based classification: *Likely Oncogenic*
(1 Strong, 1 moderate AND 2 supporting) {8 points}
- Satisfied criteria
 - OS2, gain of function demonstrated by constitutive activation and transformation in animal models (PMID: 26891877)
 - OM3, hotspot in COSMIC
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.82) and CADD prediction deleterious (score 22.7)

Notes: Allele Registry ID: CA16602568; Relevant tumor type(s): hemopoietic neoplasms; potential drug resistance to *FLT3* inhibitors (PMID: 16150941)

FLT3 p.Tyr842Cys (NM_004119.3:c.2525A>G, hg19 chr13:g.28592620T>C)

- An SOP based classification: *Likely Oncogenic*
(1 Strong, 1 moderate AND 2 supporting) {8 points}
- Satisfied criteria
 - OS2, gain of function demonstrated by constitutive activation and transformation of cultured cells (PMID: 15345593)
 - OM1, located in well-established functional domain
 - OP4, present at very low count (7) in controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.99) and CADD prediction deleterious (score 29.6)

Notes: Allele Registry ID: CA6928300; Relevant tumor type(s): hemopoietic neoplasms; potential drug resistance to *FLT3* inhibitors (PMID: 22504184, 29187377)

FLT3 p.Phe691Leu (NM_004119.3:c.2073T>G, hg19 chr13:g.28601359A>C)

- An SOP based classification: *VUS*
(1 Strong (BS1), 1 moderate AND 2 supporting) {0 points}
- Satisfied criteria
 - SBS2, F691L did not transform Ba/F3 cells (PMID: 30651561)
 - OM1, located in well-established functional domain
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.94) and CADD prediction deleterious (score 25.3)

Notes: Allele Registry ID: CA16602564; Relevant tumor type(s): hemopoietic neoplasms; potential drug resistance to *FLT3* inhibitors (PMID: 23430109, 19318574, 22504184, 29187377)

FLT3 p.Tyr693Cys (NM_004119.3:c.2078A>G, hg19 chr13:g.28601354T>C)

- An SOP based classification: *VUS*
(1 moderate AND 2 supporting) {4 points}
- Satisfied criteria
 - OM1, located in well-established functional domain
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging and CADD prediction deleterious (score 27.7)

Notes: Allele Registry ID: CA387652016; Relevant tumor type(s): hemopoietic neoplasms; potential drug resistance to *FLT3* inhibitors (PMID: 32040554)

FLT3 p.Tyr693Asn (NM_004119.3:c.2077T>A, hg19 chr13:g.28601355A>T)

- An SOP based classification: *VUS*
(1 moderate AND 2 supporting) {4 points}
- Satisfied criteria
 - OM1, located in well-established functional domain
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging and CADD prediction deleterious (score 31)

Notes: Allele Registry ID: CA387652020; Relevant tumor type(s): hemopoietic neoplasms; potential drug resistance to *FLT3* inhibitors (PMID: 32040554)

IDH1 p.Arg132His (NM_005896.4:c.395G>A, hg19 chr2:g.209113112C>T)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, gain of alternative enzymatic activity (producing 2HG) (PMID: 21446021, 19935646, 23731180)
 - OS3, hotspot in COSMIC and in cancerhotspots.org, AA position count in cancerhotspots.org is > 50 (766) and AA change count is ≥ 10 (570)
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.94) and CADD prediction deleterious (score 34)

Notes: Allele Registry ID:CA170874; Relevant tumor type(s): neoplasms

IDH1 p.Arg132Cys (NM_005896.4:c.394C>T, hg19 chr2:g.209113113G>A)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, gain of alternative enzymatic activity (producing 2HG) (PMID: 19935646, 26161668, 22885298)
 - OS3, hotspot in COSMIC and in cancerhotspots.org, AA position count in cancerhotspots.org is > 50 (766) and AA change count is ≥ 10 (139)
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.97) and CADD prediction deleterious (score 34)

Notes: Allele Registry ID:CA16602374; Relevant tumor type(s): neoplasms

IDH1 p.Arg132Ser (NM_005896.4:c.394C>A, hg19 chr2:g.209113113G>T)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, gain of alternative enzymatic activity (producing 2HG) (PMID: 19935646, 21326614)
 - OS3, hotspot in COSMIC and in cancerhotspots.org, AA position count in cancerhotspots.org is > 50 (766) and AA change count is ≥ 10 (16)
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.92) and CADD prediction deleterious (score 33)

Notes: Allele Registry ID:CA16602376; Relevant tumor type(s): neoplasms

IDH1 p.Val178Ile (NM_005896.4:c.532G>A, hg19 chr2:g.209108317C>T)

- An SOP based classification: *Benign*
(1 Very Strong) {-8 points}
- Satisfied criteria
 - SBVS1, Allele frequency > 5% in multiple populations (gnomAD v2.1.1)

Notes: Allele Registry ID: CA160063; Relevant tumor type(s): neoplasms

IDH1 p.Asp220Gly (NM_005896.4:c.659A>G, hg19 chr2:g.209108190T>C)

- An SOP based classification: *VUS*
- Satisfied criteria {0 points}

Notes: Allele Registry ID: CA160056; Relevant tumor type(s): neoplasms;
Absent from gnomAD v2.1.1 genomes, however present in 15 exomes

IDH1 p.Glu262Lys (NM_005896.4:c.784G>A, hg19 chr2:g.209106784C>T)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA350097739; Relevant tumor type(s): neoplasms

IDH1 p.Phe32Val (NM_005896.4:c.94T>G, hg19 chr2:g.209116182A>C)

- An SOP based classification: *VUS*
- Satisfied criteria {0 points}

Notes: Allele Registry ID: CA160028; Relevant tumor type(s): neoplasms;
Present in 138 exomes, 7 genomes (gnomAD v2.1.1)

IDH1 p.Val71Ile (NM_005896.4:c.211G>A, hg19 chr2:g.209113296C>T)

- An SOP based classification: *Benign*
(2 Strong AND 1 supporting) {-9 points}
- Satisfied criteria
 - SBS2, Enzymatic activity similar to wild-type IDH1 in vitro assays and does not result in increased 2HG production (PMID: 21996744, 22442146).
 - SBS1, Allele frequency > 1% (4.44%) in African population (gnomAD v2.1.1)
 - SBP1, FATHMM prediction tolerated and CADD prediction tolerated

Notes: Allele Registry ID: CA160035; Relevant tumor type(s): neoplasms

IDH1 NM_005896.4:c.123-4C>T, hg19 chr2:g.209113388G>A

- An SOP based classification: *Likely Benign*
(1 Strong AND 1 supporting) {-5 points}
- Satisfied criteria
 - SBS1, Allele frequency > 1% (2.95%) in African population (gnomAD v2.1.1)
 - SBP1, SpliceAI predicts no effect on splicing

Notes: Allele Registry ID: CA2079084; Relevant tumor type(s): neoplasms

KRAS p.Gly12Ala (NM_004985.5:c.35G>C, hg19 chr12:g.25398284C>G)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, Functional evidence PMID:26037647
 - OS3, AA change count in cancerhotspots.org is ≥ 50 (134)
 - OP4, Absent from controls
 - OP1, FATHMM prediction pathogenic (score 0.98) and CADD prediction deleterious (score 27.2)

Notes: Allele Registry ID: CA135567; Relevant tumor type(s): neoplasms.

KRAS p.Gln61Lys (NM_004985.5:c.181C>A, hg19 chr12:g.25380277G>T)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, Functional evidence PMID:28958387, 3510078
 - OS3, AA change count in cancerhotspots.org is ≥ 10 (20) and count at AA position > 50
 - OP4, Absent from controls
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 24.0)

Notes: Allele Registry ID: CA180750; Relevant tumor type(s): neoplasms.

KRAS p.Ser65Ile (NM_004985.5:c.194G>T, hg19 chr12:g.25380264C>A)

- An SOP based classification: *VUS*
(2 supporting) {1 point}
- Satisfied criteria
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 33)
 - OP4, Absent from controls

Notes: Allele Registry ID: CA384152083; In TCGA there is no tumors with p.Ser65Ile. PMID: 6822237 one child with RASopathy and de novo p.Ser65Ile. Relevant tumor type(s): malignant neoplasms.

KRAS p.Arg164Gln (NM_004985.5:c.491G>A, hg19 chr12:g.25362805C>T)

- An SOP based classification: *VUS*
(1 supporting {SBS2 downgraded} and 1 supporting {OP1}) {0 points}
- Satisfied criteria
 - SBS2 (downgraded to supporting), foci formation in NIH3T3 cells performs similarly to WT *KRAS*, however mRNA expression profiling might be indicative of slight increase in MAPK activity (PMID: [20147967](#)). Increase in nanoscale signaling complexes on the plasma membrane, however this is not well-established functional method, and in many instances, testing was done in cell line with *KRAS* p.Arg164Gln and *KRAS* p.Gly12Val (PMID: [26274561](#)).
 - OP1, FATHMM prediction pathogenic (score 0.98) and CADD prediction potentially deleterious (score 17.04)

Notes: Allele Registry ID: CA135580; Relevant tumor type(s): neoplasms.

PIK3CA p.His1047Arg (NM_006218.4:c.3140A>G, hg19 chr3:g.178952085A>G)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, Constitutive activation of the PI3K/AKT/mTOR pathway (PMID 19029981)
 - OS3, *PIK3CA* hotspot in cancerhotspots.org, AA position count in cancerhotspots.org is ≥ 50 (647) and AA change count is ≥ 10 (537)
 - OP4, Absent from controls (gnomAD v2.1.1 non-cancer)
 - OP1, FATHMM prediction pathogenic (score 0.96) and CADD prediction likely deleterious (score 22.5)

Notes: Allele Registry ID: CA123326; Relevant tumor type(s): neoplasms.

PIK3CA p.His1047Gln (NM_006218.4:c.3141T>G, hg19 chr3:g.178952086T>G)

- An SOP based classification: *VUS*
(1 moderate AND 2 supporting) {4 points}
- Satisfied criteria
 - OM4, Missense change at an amino acid residue where a different missense change determined to be oncogenic/pathogenic has been documented.
 - OP3, *PIK3CA* hotspot in cancerhotspots.org, AA change count is < 10 (7)
 - OP4, Absent from controls (gnomAD)

Notes: Allele Registry ID: CA16602915; Relevant tumor type(s): neoplasms.

CADD prediction uncertain (score 6.76)

PIK3CA p.His1065Tyr (NM_006218.4:c.3193C>T, hg19 chr3:g.178952138C>T)

- An SOP based classification: *VUS*
(3 supporting) {3 points}
- Satisfied criteria
 - OP1, FATHMM prediction pathogenic (score 0.94) and CADD prediction potentially deleterious (score 15.57)
 - OP3, *PIK3CA* hotspot in cancerhotspots.org, AA change count is < 10 (4)
 - OP4, Absent from controls (gnomAD)

Notes: Allele Registry ID: CA355286031; Relevant tumor type(s): neoplasms.

PIK3CA p.Lys733Arg (NM_006218.4:c.2198A>G, hg19 chr3:g.178941879A>G)

- An SOP based classification: *VUS*
(1 moderate) {2 points}
- Satisfied criteria
 - OP1, FATHMM prediction pathogenic (score 0.97) and CADD prediction likely deleterious (score 22.4)

Notes: Allele Registry ID: CA2710903; Relevant tumor type(s): neoplasms.

Allele frequency of 0.0056 (0.56%, 110/19,494 alleles) in the East Asian gnomAD cohort.

PIK3CA p.Glu545Ala (NM_006218.4:c.1634A>C, hg19 chr3:g.178936092A>C)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, Increased phosphorylation of Akt and transformation in cell culture (PMID: 17376864, 29533785)
 - OS3, *PIK3CA* hotspot in cancerhotspots.org, AA position count in cancerhotspots.org is ≥ 50 (633) and AA change count is ≥ 10 (12)
 - OP4, Absent from controls (gnomAD)
 - OP1, FATHMM prediction pathogenic (score 0.97) and CADD prediction deleterious (score 27.6)

Notes: Allele Registry ID: CA123342; Relevant tumor type(s): neoplasms.

PIK3CA p.Glu453Lys (NM_006218.4:c.1357G>A, hg19 chr3:g.178928079G>A)

- An SOP based classification: *Likely Oncogenic*
(1 Strong AND 1 moderate AND 2 supporting) {8 points}
- Satisfied criteria
 - OS2, Increased phosphorylation of Akt and Mek1/2, enhanced cell survival, and transforming in culture (PMID: 26627007, 29533785)
 - OM3, *PIK3CA* hotspot in cancerhotspots.org, AA position count in cancerhotspots.org is < 50 (42) and AA change count is ≥ 10 (31)
 - OP4, Absent from controls (gnomAD)
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 32)

Notes: Allele Registry ID: CA16602904; Relevant tumor type(s): neoplasms.

PIK3CA p.Glu418Lys (NM_006218.4:c.1252G>A, hg19 chr3:g.178927974G>A)

- An SOP based classification: *VUS*
(2 supporting) {2 points}
- Satisfied criteria
 - OP4, Absent from controls (gnomAD)
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 25.9)

Notes: Allele Registry ID: CA136365; Relevant tumor type(s): neoplasms.

PIK3CA p.Ile391Met (NM_006218.4:c.1173A>G, hg19 chr3:g.178927410A>G)

- An SOP based classification: *Benign*
(1 Very Strong) {-8 points}
- Satisfied criteria
 - SBVS1, Allele frequency of 0.066 (6.6%, 18372/279610 alleles) in gnomAD

Notes: Allele Registry ID: CA161490; Relevant tumor type(s): neoplasms.

PIK3CA p.Met1? (NM_006218.4:c.1A>G, hg19 chr3:g.178916614A>G)

- An SOP based classification: *VUS*
(3 supporting) {3 points}
- Satisfied criteria
 - OP1, FATHMM prediction pathogenic (score 1.00) and CADD prediction deleterious (score 23.7)
 - OP3, *PIK3CA* hotspot in cancerhotspots.org, AA change count is < 10 (6)
 - OP4, Absent from controls (gnomAD)

Notes: Allele Registry ID: CA355269996; Relevant tumor type(s): neoplasms.

PIK3CA p.Asn1068LysfsTer5 (NM_006218.4:c.3203dup, hg19 chr3:g.178952148dup)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, Absent from controls (gnomAD)

Notes: Allele Registry ID: CA123341; Stop codon loss; Relevant tumor type(s): neoplasms.
PMID: 21703185 seems to suggest lack of activation of Akt/mTor signaling.

PTEN p.Arg130Gln (NM_000314.6:c.389G>A, hg19 chr10:g.89692905G>A)
(*PTEN* germline Expert Panel: Pathogenic)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, Phosphatase activity <50% of wild type (PMID 10866302)
 - OS3, *PTEN* hotspot in cancerhotspots.org, AA position count in cancerhotspots.org is $\geq$ 50 (168) and AA change count is $\geq$ 10 (72)
 - OP4, Absent from controls (gnomAD)
 - OP1, FATHMM prediction pathogenic (score 0.99) and CADD prediction deleterious (score 29.8)

Notes: Allele Registry ID: CA000437; Relevant tumor type(s): neoplasms.

PTEN p.Thr131Ile (NM_000314.6:c.392C>T, hg19 chr10:g.89692908C>T)
(*PTEN* germline Expert Panel: Pathogenic)

- An SOP based classification: *Likely Oncogenic*
(1 Strong AND 1 moderate AND 1 supporting) {7 points}
- Satisfied criteria
 - OS2, Phosphatase activity <50% of wild type (PMID 21828076)
 - OM1, Located in well-established functional domain (phosphatase domain)
 - OP4, Absent from controls (gnomAD)

Notes: Allele Registry ID: CA000442; Relevant tumor type(s): neoplasms.

Probands with phenotype specificity score of 2-3.5. (PMID 23160955, 23335809)

PTEN p.Lys322Ter (NM_000314.6:c.964A>T, hg19 chr10:g.89720813A>T)
(*PTEN* germline Expert Panel: Likely Pathogenic)

- An SOP based classification: *Likely Oncogenic*
(1 Very Strong AND 1 supporting) {9 points}
- Satisfied criteria
 - OVS1, nonsense variant, NMD predicted
 - OP4, Absent from controls

Notes: Allele Registry ID: CA000235; Relevant tumor type(s): neoplasms.

PTEN p.His123Arg (NM_000314.6:c.368A>G, hg19 chr10:g.89692884A>G) (*PTEN* germline Expert Panel: Pathogenic)

- An SOP based classification: *Likely Oncogenic*
(1 strong, 1 moderate AND 2 supporting) {7 points}
- Satisfied criteria
 - OS2, Phosphatase activity <50% of wild type (PMID 29706350)
 - OM1, Located in well-established functional domain (phosphatase domain)
 - OP4, Absent from controls (gnomAD)
 - OP1, FATHMM prediction pathogenic (score 0.98) and CADD prediction likely deleterious (score 24.5)

Notes: Allele Registry ID: CA000418; Relevant tumor type(s): neoplasms.

Assumed de novo, but without confirmation of paternity and maternity in a patient with the disease and no family history. (PMID 10234502)

PTEN NM_000314.6:c.634+5G>C (hg19 chr10:g.89712021G>C)
(*PTEN* germline Expert Panel: Likely Pathogenic)

- An SOP based classification: *Likely Oncogenic*
(1 Strong AND 2 supporting) {6 points}
- Satisfied criteria
 - OVS1 (downgraded to Strong), Variant shown to cause exon 6 skipping (PMID 28677221)
 - OP4, Absent from controls (gnomAD)
 - OP1, SpliceAI and varSEAK predict splicing defects

Notes: Allele Registry ID: CA645509438; Relevant tumor type(s): neoplasms.

PTEN p.Pro204Ala (NM_000314.6:c.610C>G, hg19 chr10:g.89711992C>G)
(*PTEN* germline Expert Panel: **Likely Pathogenic**)

- An SOP based classification: **VUS**
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, Absent from controls (gnomAD)

Notes: Allele Registry ID: CA000535; Relevant tumor type(s): neoplasms.

Assumed de novo, but without confirmation of paternity and maternity in a patient with the disease and no family history. (internal laboratory contributor SCV000222125.5)

Proband(s) with phenotype specificity score of 1-1.5. (internal laboratory contributor SCV000222125.5)

PTEN p.Leu182Ser (NM_000314.6:c.545T>C, hg19 chr10:g.89711927T>C)
(*PTEN* germline Expert Panel: VUS)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, Absent from controls (gnomAD)

Notes: Allele Registry ID: CA000177; Relevant tumor type(s): neoplasms.

PTEN p.Pro354Leu (NM_000314.6:c.1061C>T, hg19 chr10:g.89725078C>T)
(*PTEN* germline Expert Panel: VUS)

- An SOP based classification: *VUS*
(1 supporting {OP4} AND 1 supporting {SBS2 downgraded}) {0 points}
- Satisfied criteria
 - SBS2 {downgraded to supporting}, Functional studies showing potentially no damaging effect on protein function, but data is not strong enough. (PMID 29706350)
 - OP4, Very low in controls (gnomAD)

Notes: Allele Registry ID: CA000277; Relevant tumor type(s): neoplasms.

PTEN p.Leu193= (NM_000314.6:c.579G>A, hg19 chr10:g.89711961G>A)
(*PTEN* germline Expert Panel: Likely Benign)

- An SOP based classification: *Likely Benign*
(1 Strong AND 1 supporting) {-5 points}
- Satisfied criteria
 - SBS1, Allele frequency of 0.00125 (0.125%, 13/10,360 alleles) in the Ashkenazi Jewish subpopulation of the gnomAD cohort.
 - SBP2, A synonymous variant for which splicing prediction algorithms SpliceAI and varSEAK predict no impact on splicing.

Notes: Allele Registry ID: CA000523; Relevant tumor type(s): neoplasms.

PTEN NM_000314.6:c.-9C>G (hg19 chr10:g.89624218C>G)
(*PTEN* germline Expert Panel: Benign)

- An SOP based classification: *Benign*
(1 Very Strong) {-8 points}
- Satisfied criteria
 - SBVS1, Allele frequency of 0.048 (4.8%, 903/18,870 alleles) in the East Asian subpopulation of the gnomAD cohort.

Notes: Allele Registry ID: CA000663; Relevant tumor type(s): neoplasms.

RB1 p.Ile124ArgfsTer6 (NM_000321.2:c.371_372del, hg19 chr13:g.48916841_48916842del)

- An SOP based classification: *Oncogenic*
(1 Very Strong AND 1 moderate AND 1 supporting) {11 points}
- Satisfied criteria
 - OVS1, frameshift in tumor suppressor gene (Results in a premature stop codon 6 amino acids downstream of AA 124, NMD predicted)
 - OM1, loss of 3 functional domains
 - OP4, absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA645578601; Relevant tumor type(s): neoplasms

***RB1* p.Ser443Ter (NM_000321.2:c.1328C>A , hg19 chr13:g.48951166C>A)**

- An SOP based classification: *Oncogenic*
(1 Very Strong AND 1 moderate AND 1 supporting) {11 points}
- Satisfied criteria
 - OVS1, premature stop codon in tumor suppressor gene, NMD predicted
 - OM1, loss of 2 functional domains
 - OP4, absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA16614041; Relevant tumor type(s): neoplasms;

Variant has been reported in an individual with retinoblastoma (PMID: 12541220)

RB1 NM_000321.2:c.607+1G>A (hg19 chr13:g.48923160G>A)

- An SOP based classification: *Likely Oncogenic*
(1 Strong AND 2 supporting) {6 points}
- Satisfied criteria
 - OVS1 (downgraded to Strong), Canonical ± 1 or 2 splice site in tumor suppressor gene, however potential exon skipping without frameshift
 - OP1, SpliceAI and varSEAK predict splicing defects
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA16619811; Relevant tumor type(s): neoplasms;

Reported in several individuals with either a personal or family history of retinoblastoma (PMID: 16343894, 14722923, 12541220)

RB1 NM_000321.2:c.2211+5G>T (hg19 chr13:g.49037976G>T)

- An SOP based classification: *Likely Oncogenic*
(1 Strong AND 2 supporting) {6 points}
- Satisfied criteria
 - OVS1 (downgraded to Strong), Exon 21 skipping, based on review of RNA-seq in NCI-H1436 cancer cell line which has *RB1* NM_000321.2:c.2211+5G>T variant in MutSpliceDB, however exon skipping without frameshift, exon in important domain (https://brb.nci.nih.gov/cgi-bin/splicing/splicing_evidence.cgi?caid=CA658760346)
 - OP1, SpliceAI and varSEAK predict splicing defects
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA658760346; Relevant tumor type(s): neoplasms

RB1 NM_000321.2:c.1216-29A>G, hg19 chr13:g.48951025A>G)

- An SOP based classification: *Benign*
(1 Very Strong) {-8 points}
- Satisfied criteria
 - SBVS1, Allele frequency of 0.083 (8.3%, 2083/24904 alleles) in gnomAD (African population)

Notes: Allele Registry ID: CA027882; Relevant tumor type(s): neoplasms

RB1 p.Gln689His (NM_000321.2:c.2067G>C, hg19 chr13:g.49033930G>C)

- An SOP based classification: *VUS*
(1 moderate AND 1 supporting) {3 points}
- Satisfied criteria
 - OM1, Located in *RB1* B domain
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA388166885; Relevant tumor type(s): neoplasms

RB1 p.Ser567Leu (NM_000321.2:c.1700C>T, hg19 chr13:g.49027133C>T)

- An SOP based classification: *VUS*
(1 moderate AND 2 supporting) {4 points}
- Satisfied criteria
 - OM1, Located in *RB1* A domain
 - OP4, Absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.91) and CADD prediction deleterious (score 34)

Notes: Allele Registry ID: CA026396; Relevant tumor type(s): neoplasms;

Present in individual with hereditary retinoblastoma, however not clear if it's present with or without p.Cys712Arg (PMID: 10671068), p.Cys712Arg has functional data suggesting loss of function.

RB1 p.Trp563Leu (NM_000321.2:c.1688G>T, hg19 chr13:g.48955572G>T)

- An SOP based classification: *VUS*
(1 moderate AND 1 supporting) {3 points}
- Satisfied criteria
 - OM1, Located in *RB1* A domain
 - OP4, Absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA388164056; Relevant tumor type(s): neoplasms

RB1 p.Arg661Trp (NM_000321.2:c.1981C>T, hg19 chr13:g.49033844C>T)

- An SOP based classification: *Likely Oncogenic* (2 moderate AND 2 supporting) {6 points}
- Satisfied criteria
 - OS2 (downgraded to moderate), partial loss of function (PMID: 18677112, 18682685, 10486322, 16449662, 15643604, 9632788)
 - OM1, located in *RB1* B domain
 - OP4, present at very low numbers (4) in controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.96) and CADD prediction deleterious (score 35)

Notes: Allele Registry ID: CA026417; Relevant tumor type(s): neoplasms;

Variant segregate with retinoblastoma in multiple families (PMID: 1352883, 26925970, 17096365), however penetrance was reduced in comparison to truncating variants. Perhaps this is a risk allele.

RB1 p.Glu554Lys (NM_000321.2:c.1660G>A, hg19 chr13:g.48955544G>A)

- An SOP based classification: *VUS*
(1 moderate AND 2 supporting) {3 points}
- Satisfied criteria
 - OM1, located in *RB1* A domain
 - OP4, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.99) and CADD prediction deleterious (score 33)

Notes: Allele Registry ID: CA388163960; Relevant tumor type(s): neoplasms

RB1 p.Tyr606Cys (NM_000321.2:c.1817A>G, hg19 chr13:g.49030342A>G)

- An SOP based classification: *VUS*
(1 moderate AND 2 supporting) {4 points}
- Satisfied criteria
 - OM2, absent from controls (gnomAD v2.1.1)
 - OP1, FATHMM prediction damaging (score 0.99) and CADD prediction deleterious (score 27.5)
 - OP3, hotspot in cancerhotspots.org, AA position count in cancerhotspots.org is < 10 (2) and AA change count is < 10 (2)

Notes: Allele Registry ID: CA388165692; Relevant tumor type(s): neoplasms; Three bp from exon-intron junction.

TERT hg19 chr5:g.1295228G>A (c.-124C>T)

- An SOP based classification: *Likely Oncogenic*
(2 Strong, 1 supporting) {9 points}
- Satisfied criteria
 - OS2, increased *TERT* expression (PMID: 23348506, 23348503)
 - OS3, hotspot (PMID: 23348506, 23348503, 23530248, 31408918)
 - OP4, very low count in gnomAD v2.1.1 (3)

Notes: Allele Registry ID: CA557858711; Relevant tumor type(s): neoplasms

TERT hg19 chr5:g.1295250G>A (c.-146C>T)

- An SOP based classification: *Likely Oncogenic*
(2 Strong, 1 supporting) {9 points}
- Satisfied criteria
 - OS2, increased *TERT* expression (PMID: 23348506, 23348503)
 - OS3, hotspot (PMID: 23348506, 23348503, 23530248, 31408918)
 - OP4, absent in gnomAD v2.1.1

Notes: Allele Registry ID: CA649878366; Relevant tumor type(s): neoplasms

TERT hg19 chr5:g.1295228G>T (c.-124C>A)

- An SOP based classification: *VUS*
(1 Strong, 1 supporting) {5 points}
- Satisfied criteria
 - OS2, increased *TERT* expression (PMID: 31408918)
 - OP4, absent in gnomAD v2.1.1 (3)

Notes: Allele Registry ID: CA649878362; Relevant tumor type(s): neoplasms

TERT hg19 chr5:g.1295436G>A (c.-332C>T)

- An SOP based classification: *VUS*
(1 Strong, 1 supporting) {5 points}
- Satisfied criteria
 - OS2, increased *TERT* expression (PMID: 31408918)
 - OP4, absent in gnomAD v2.1.1 (3)

Notes: Allele Registry ID: CA923726366; Relevant tumor type(s): neoplasms

TERT hg19 chr5:g.1295349A>G (c.-245T>C)

- An SOP based classification: *Benign*
(1 Very Strong) {-8 points}
- Satisfied criteria
 - SBVS1, Minor allele frequency is >5%, 24% in gnomAD v2.1.1

Notes: Allele Registry ID: CA11957810; Relevant tumor type(s): neoplasms

TERT p.Arg230Ter (NM_198253.3:c.688C>T, hg19 chr5:g.1294313G>A)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA359057024; Relevant tumor type(s): neoplasms

TERT p.Trp581Ter (NM_198253.3:c.1743G>A, hg19 chr5:g.1282570C>T)

- An SOP based classification: *VUS*
(1 supporting) {1 point}
- Satisfied criteria
 - OP4, absent from controls (gnomAD v2.1.1)

Notes: Allele Registry ID: CA359082862; Relevant tumor type(s): neoplasms

This nonsense change has been observed in individual(s) with dyskeratosis congenita (PMID: 26024875).

TP53 p.Val173Met (NM_000546.5:c.517G>A, hg19 chr17:g.7578413C>T) (*TP53* germline Expert Panel: Pathogenic)

- An SOP based classification: *Oncogenic*
(2 Strong AND 2 supporting) {10 points}
- Satisfied criteria
 - OS2, Median IARC transactivation activity < 20 and p53WTNutlin3 Z-score >= 0.61 and Etoposide Z-score <= -0.21
 - OS3, *TP53* hotspot in cancerhotspots.org, AA position count in cancerhotspots.org is ≥ 50 (78) and AA change count is ≥ 10 (27)
 - OP4, Absent from controls (gnomAD)
 - OP1, BayesDel > 0.16 and AGVGD Class > C15

Notes: Allele Registry ID: CA10577579; Relevant tumor type(s): neoplasms.

This variant has been reported in a proband meeting classic Li-Fraumeni syndrome criteria (PMID: 16494995). There is a de novo observation of a proband with a diagnosis of a sarcoma, breast cancer and thyroid cancers under the age of 26 years with parental confirmation (ClinVar SCV000545293.4).

TP53 p.Cys124Ter (NM_000546.5:c.372C>A, hg19 chr17:g.7579315G>T) (*TP53* germline Expert Panel: Pathogenic)

- An SOP based classification: *Oncogenic*
(1 Very Strong AND 1 moderate AND 1 supporting) {11 points}
- Satisfied criteria
 - OVS1, nonsense variant in a bona fide tumor suppressor gene, NMD predicted
 - OM1, located in well-established functional domain (DNA binding domain)
 - OP4, absent from controls (gnomAD)

Notes: Allele Registry ID: CA397844214; Relevant tumor type(s): neoplasms.

There is a de novo observation of a proband with adrenocortical carcinoma in childhood without mention of parental confirmation (PMID: 22233476).

TP53 p.Cys135Gly (NM_000546.5:c.403T>G, hg19 chr17:g.7578527A>C) (*TP53* germline Expert Panel: Likely Pathogenic)

- An SOP based classification: *Likely Oncogenic*
(1 Strong, 1 moderate AND 3 supporting) {9 points}
- Satisfied criteria
 - OS2, Median IARC transactivation activity < 20 and p53WTNutlin3 Z-score >= 0.61 and Etoposide Z-score <= -0.21
 - OM1, Located in well-established functional domain (DNA binding domain)
 - OP4: Absent from controls (gnomAD)
 - OP3, *TP53* hotspot in cancerhotspots.org, AA change count in cancerhotspots.org is < 10 (2)
 - OP1, BayesDel > 0.16 and AGVGD Class > C15

Notes: Allele Registry ID: CA16602989; Relevant tumor type(s): neoplasms.

TP53 p.Glu180Lys (NM_000546.5:c.538G>A, hg19 chr17:g.7578392C>T) (*TP53* germline Expert Panel: Likely Pathogenic)

- An SOP based classification: *Likely Oncogenic*
(2 moderate AND 2 supporting) {6 points}
- Satisfied criteria
 - OS2 (downgraded to moderate), Median IARC transactivation activity between 20%-75% and p53WTNutlin3 Z-score ≥ 0.61 and Etoposide Z-score ≤ -0.21
 - OM1, Located in well-established functional domain (DNA binding domain)
 - OP4: Absent from controls (gnomAD)
 - OP1, BayesDel > 0.16 and AGVGD Class $> C15$

Notes: Allele Registry ID: CA10584590; Relevant tumor type(s): neoplasms.

This variant has been reported in 2 probands meeting Chompret criteria (PMID: 8118819, 25584008)

TP53 p.Ala347Asp (NM_000546.5:c.1040C>A, hg19 chr17:g.7573987G>T) (*TP53* germline Expert Panel: Likely Pathogenic)

- An SOP based classification: *Likely Oncogenic*
(1 Strong AND 1 moderate AND 1 supporting) {7 points}
- Satisfied criteria
 - OS2, Median IARC transactivation activity < 20 and a tetramerization assay demonstrates that a mutant protein does not form tetramers (PMID: 12826609, PMID: 16007150).
 - OM1, Located in well-established functional domain (tetramerization domain)
 - OP4, Absent from controls (gnomAD)

Notes: Allele Registry ID: CA000022; Relevant tumor type(s): neoplasms.

This variant has been reported in 1 proband meeting classic Li-Fraumeni syndrome criteria (PMID: 27496084). The variant was found to co-segregate with disease in this proband's family, with 5 or 6 meioses observed (PMID: 27496084).

TP53 p.Pro47Ser (NM_000546.5:c.139C>T, hg19 chr17:g.7579548G>A) (*TP53* germline Expert Panel: Benign)

- An SOP based classification: *Benign*
(1 Very Strong, 1 Strong AND 1 supporting) {-13 points}
- Satisfied criteria
 - SBVS1, Minor allele frequency is > 0.1% (*TP53* EP cut off)
(1.6%, 410/24922 alleles in the African subpopulation of the gnomAD)
 - SBS2, Median IARC transactivation activity > 75% and p53WTNutlin3 Z-score < 0.61 and Etoposide Z-score > -0.21
 - SBP1, BayesDel < 0.16 and AGVGD Class < C15

Notes: Allele Registry ID: CA000053; Relevant tumor type(s): neoplasms.

This variant has been observed in at least 105 60+ year old females without a cancer diagnosis (FLOSSIES database - <https://whi.color.com>).

TP53 p.Val73Met (NM_000546.5:c.217G>A, hg19 chr17:g.7579470C>T) (*TP53* germline Expert Panel: Benign)

- An SOP based classification: *Benign*
(2 Strong AND 1 supporting) {-9 points}
- Satisfied criteria
 - SBS1, Minor allele frequency is > 0.03% (TP53 EP cut off)
(0.05%, 13/25106 alleles in the European (Finnish) subpopulation of the gnomAD)
 - SBS2, Median IARC transactivation activity > 75% and p53WTNutlin3 Z-score < 0.61 and Etoposide Z-score > -0.21
 - SBP1, BayesDel < 0.16 and AGVGD Class < C15

Notes: Allele Registry ID: CA000075; Relevant tumor type(s): neoplasms.

***TP53* p.His365Tyr (NM_000546.5:c.1093C>T, hg19 chr17:g.7573934G>A)
(*TP53* germline Expert Panel: Likely Benign)**

- An SOP based classification: *Likely Benign*
(1 Strong AND 1 supporting) {-5 points}
- Satisfied criteria
 - SBS2, Median IARC transactivation activity > 75% and p53WTNutlin3 Z-score < 0.61 and Etoposide Z-score > -0.21
 - SBP1, BayesDel < 0.16 and AGVGD Class < C15

Notes: Allele Registry ID: CA000035; Relevant tumor type(s): neoplasms.

TP53 p.Glu298Lys (NM_000546.5:c.892G>A, hg19 chr17:g.7577046C>T)
(*TP53* germline Expert Panel: Likely Benign)

- An SOP based classification: *Likely Benign*
(1 Strong AND 1 supporting) {-5 points}
- Satisfied criteria
 - SBS2, Median IARC transactivation activity > 75% and p53WTNutlin3 Z-score < 0.61 and Etoposide Z-score > -0.21
 - SBP1, BayesDel < 0.16 and AGVGD Class < C15

Notes: Allele Registry ID: CA000483; Relevant tumor type(s): neoplasms.

TP53 p.Gly334Arg (NM_000546.5:c.1000G>C, hg19 chr17:g.7574027C>G) (*TP53* germline Expert Panel: VUS)

- An SOP based classification: *VUS*
(1 Strong {BS1}, 1 moderate OM1 AND 2 supporting {OP1, OP4}) {0 points}
- Satisfied criteria
 - SBS2, Median IARC transactivation activity > 75% and p53WTNutlin3 Z-score < 0.61 and Etoposide Z-score > -0.21
 - OM1, Located in well-established functional domain (DNA binding domain)
 - OP4, Absent from controls (gnomAD)
 - OP1, BayesDel > 0.16 and AGVGD Class > C15

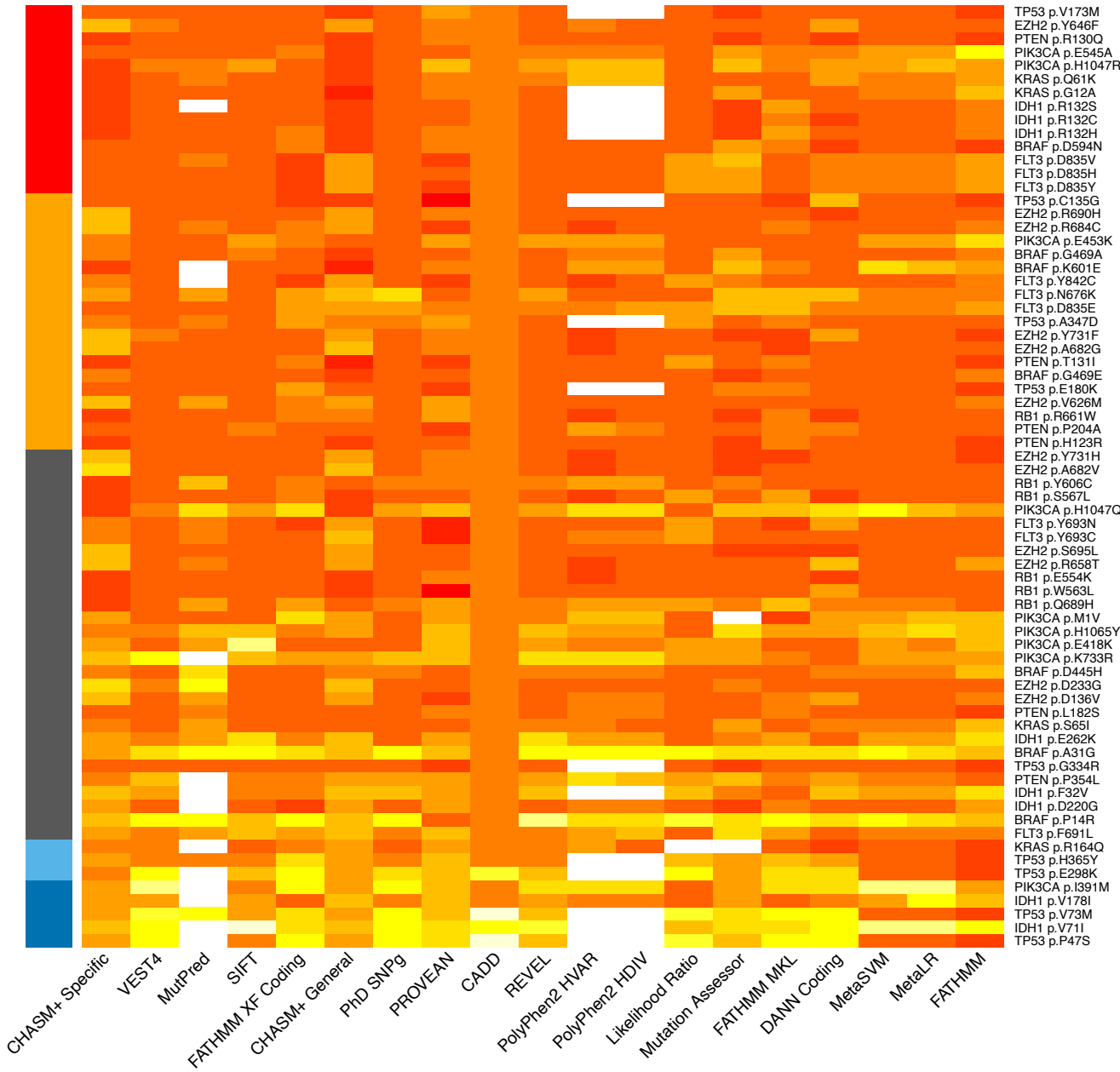
Notes: Allele Registry ID: CA000007; Relevant tumor type(s): neoplasms.

This variant has been reported in 2 probands meeting Chompret criteria (PMID: 23580068, 25584008).

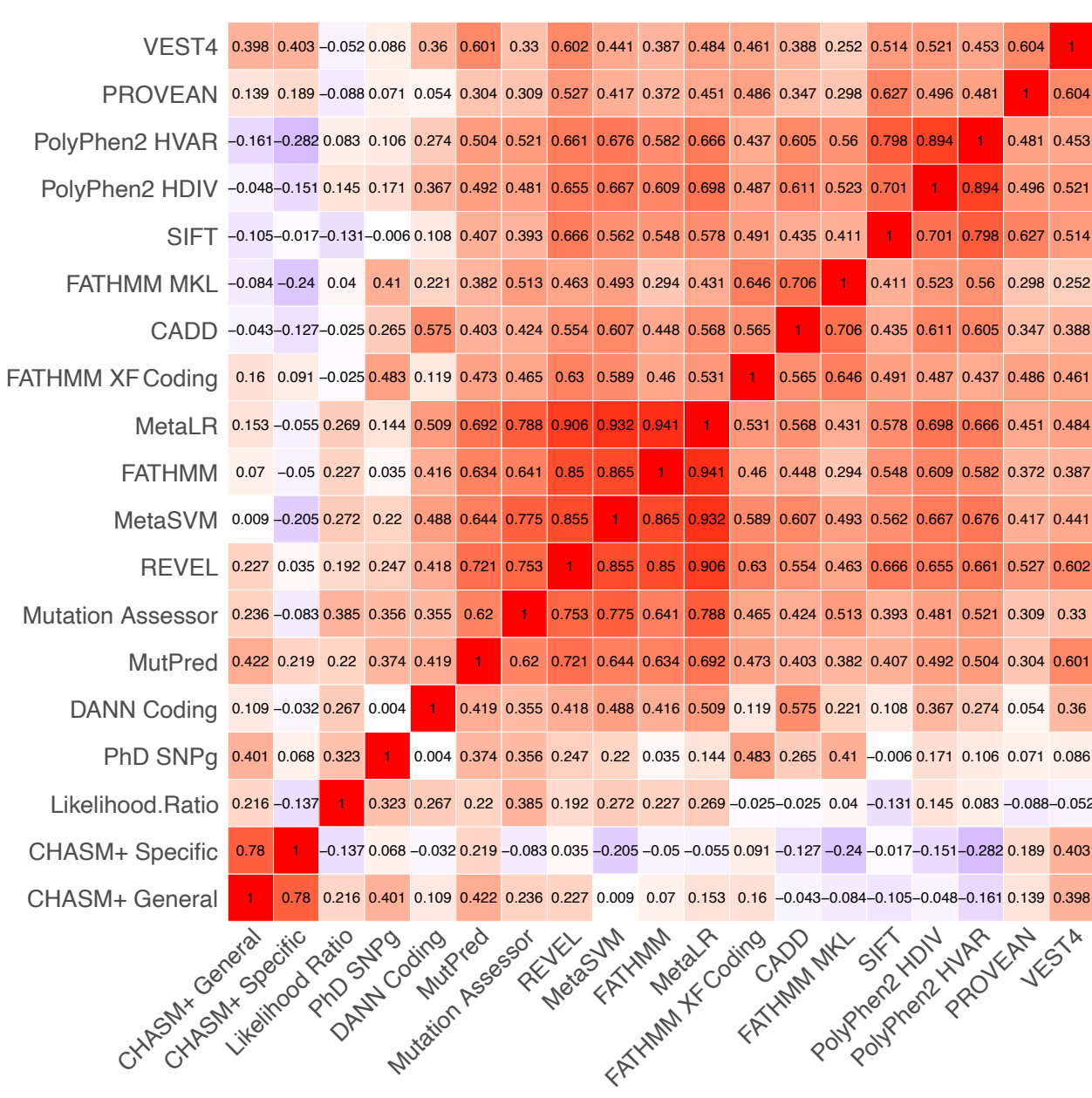
Supplementary Methods

We performed an analysis of 19 algorithms for in silico prediction of variant effect for a subset of the test variants used to demonstrate the Oncogenicity SOP. Specifically, this analysis was limited to the 70 variants of Table 1 that are single nucleotide missense variants occurring within the coding region of an exon. For assessing algorithm performance, variants achieving a classification of Oncogenic or Likely Oncogenic were considered Oncogenic and variants with a classification of Benign or Likely Benign were considered to be not oncogenic. Scores for each algorithm were obtained using OpenCRAVAT (PMID: 32228266) where possible. Scores for the PROVEAN algorithm were obtained using an additional web resource dedicated to this algorithm. To facilitate interpretation, scores from some algorithms were converted to a scale of 0 to 1 and/or inverted so that high values indicated greater predicted variant effect. CADD phred scores were converted to p-values by the following calculation: $10^{-(\text{Score}/10)}$ and these p-values were then subtracted from 1. VEST-4 p-values were subtracted from 1. The PROVEAN scores were rescaled to 0 to 1 by the following calculation: $(\text{Score} - \text{Min Score}) / (\text{Max Score} - \text{Min Score})$. The re-scaled scores were then subtracted from 1. Spearman correlations were calculated to compare each algorithm's scores to the Oncogenicity SOP points. Spearman correlations were also used to compare each set of algorithm scores to every other algorithm. A receiver operator characteristic (ROC) area under the curve (AUC) analysis was performed to assess the general performance of each algorithm's score in discriminating oncogenic from benign variants. Variants of Uncertain Significance (VUS) were ignored for the ROC AUC analysis. ROC AUC calculation was performed using the "pROC" package in R. The same set of Oncogenic and Benign variants defined according to the Oncogenicity SOP classification were used to assess Positive Predictive Value (PPV) and Negative Predictive Value (NPV) when compared to each algorithm's own classification using score cutoffs recommended by the developer of each tool. Many of these algorithms have their own scheme for designating oncogenic/pathogenic variants. For the CADD algorithm, "Deleterious" was considered equivalent to Oncogenic Support, and "Tolerated" was considered equivalent to Benign Support. For the FATHMM, MetaLR, MetaSVM and SIFT algorithms, "Damaging" was considered equivalent to Oncogenic Support, and "Tolerated" was considered equivalent to Benign Support. For the FATHMM MKL, FATHMM XF Coding and Likelihood Ratio Test algorithms, "Damaging" was considered equivalent to Oncogenic Support, and "Neutral" was considered equivalent to Benign Support. For the Mutation Assessor algorithm, "High/Medium" was considered equivalent to Oncogenic Support, and "Low/Neutral" was considered equivalent to Benign Support. For the PhD-SNPg algorithm, "Pathogenic" was considered equivalent to Oncogenic Support, and "Benign" was considered equivalent to Benign Support. For the Polyphen HDIV and Polyphen HVAR algorithms, "D/P" was considered equivalent to Oncogenic Support, and "B" was considered equivalent to Benign Support. For the PROVEAN algorithm "Deleterious" was considered equivalent to Oncogenic Support, and "Neutral" was considered equivalent to Benign Support. For the REVEL algorithm, "Likely disease causing" was considered equivalent to Oncogenic Support, and "Likely benign" was considered equivalent to Benign Support. For the VEST4 algorithm, "Pathogenic" was considered equivalent to Oncogenic Support, and "Benign" was considered equivalent to Benign Support.

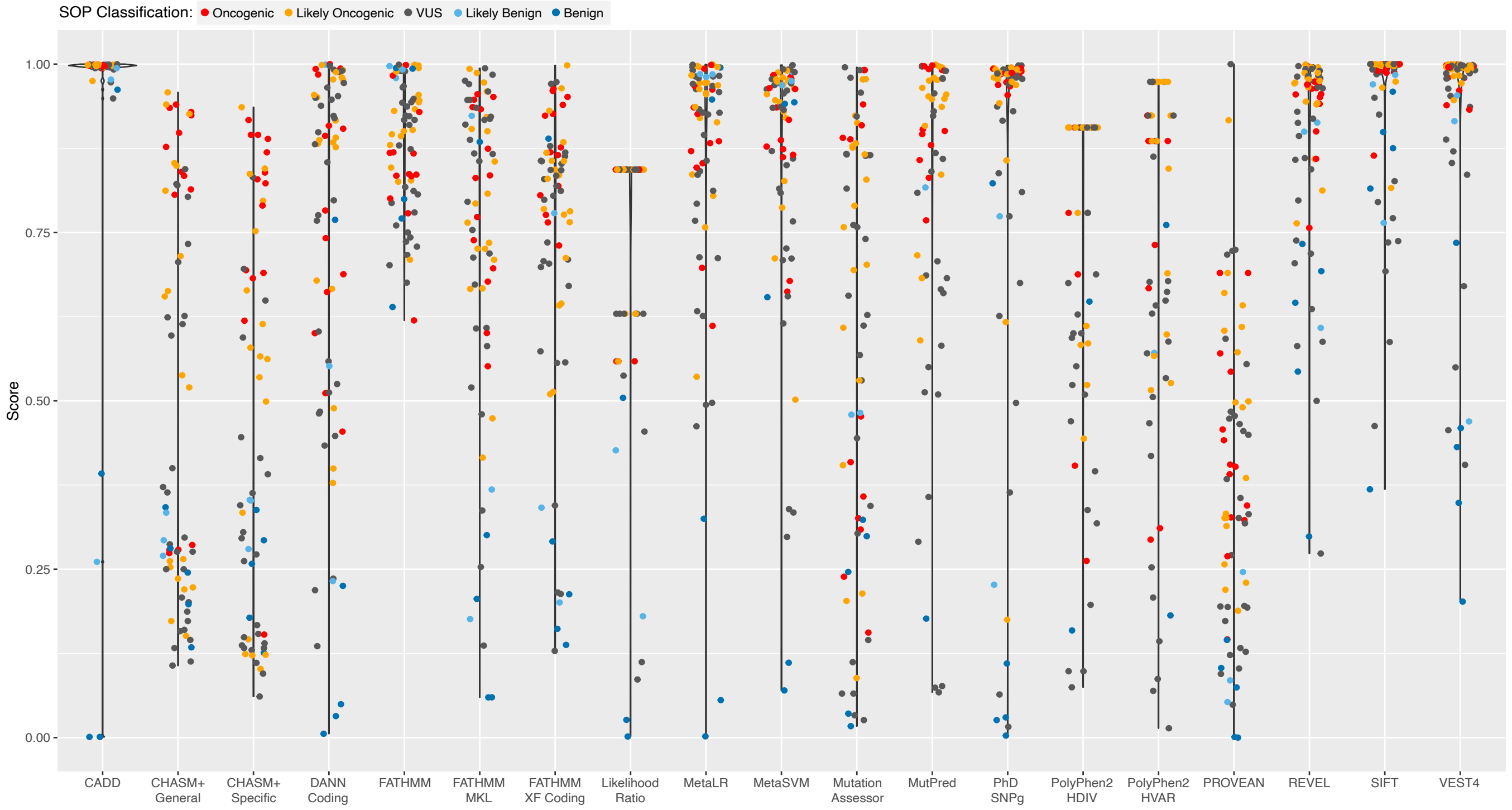
A



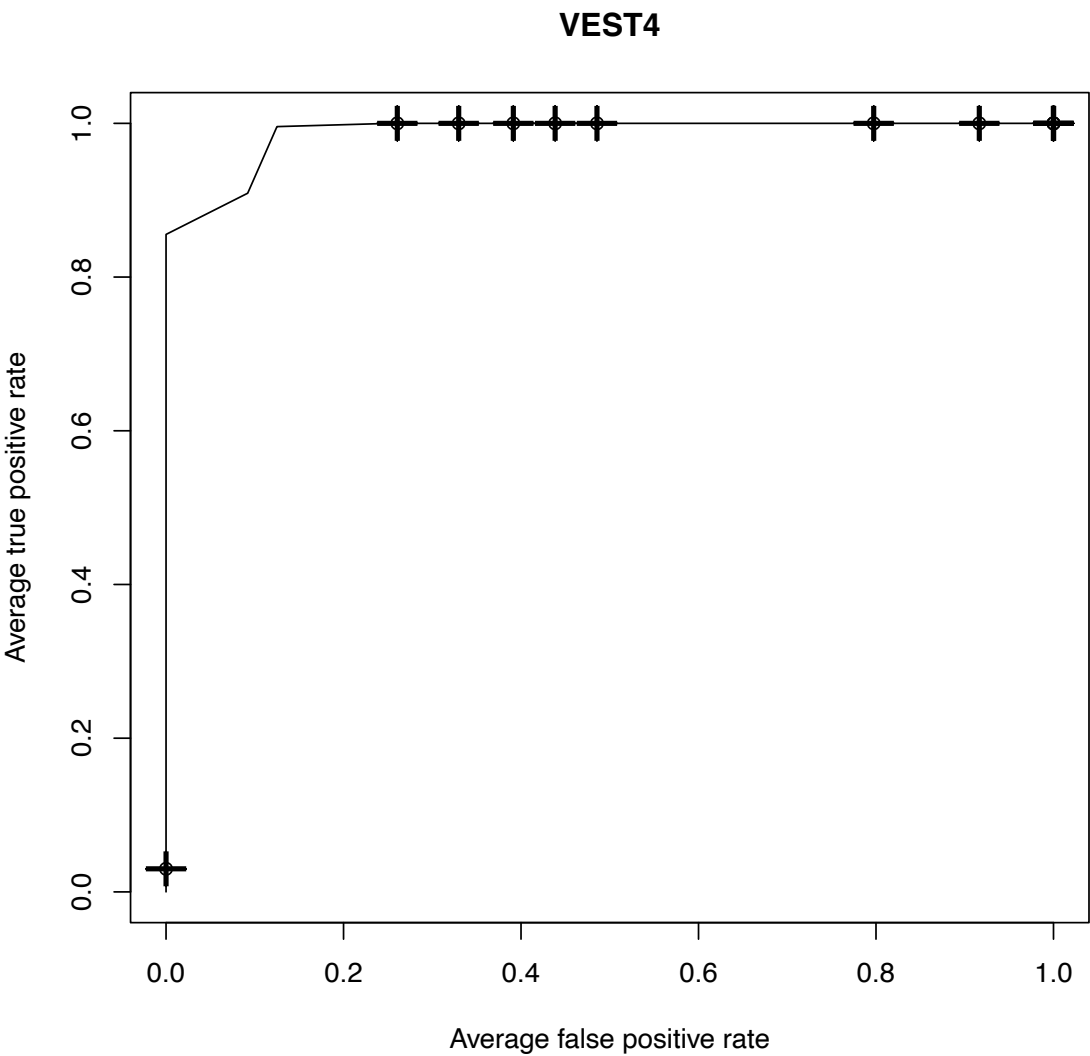
B



C



D



E

Prediction Algorithm	Correlation	PPV	NPV	ROC AUC
VEST4	0.531	0.917	1.000	0.989
PROVEAN	0.450	0.967	0.6	0.985
CADD	0.435	0.917	1.000	0.981
REVEL	0.424	N/A	N/A	0.958
SIFT	0.496	0.912	0.857	0.951
MutPred	0.530	N/A	N/A	0.933
PhD SNPg	0.456	0.914	0.833	0.915
FATHMM XF Coding	0.475	0.939	0.750	0.875
Mutation Assessor	0.297	1.000	0.412	0.835
FATHMM MKL	0.284	0.892	1.000	0.832
Likelihood Ratio	0.316	0.941	0.833	0.827
DANN Coding	0.280	N/A	N/A	0.816
PolyPhen2 HVAR	0.421	0.920	0.333	0.813
CHASM+ Specific	0.561	N/A	N/A	0.809
PolyPhen2 HDIV	0.424	0.920	0.500	0.780
CHASM+ General	0.467	N/A	N/A	0.765
MetaSVM	0.142	0.875	0.375	0.591
MetaLR	0.127	0.848	0.375	0.545
FATHMM	0.035	0.853	0.429	0.469
MEAN	0.376	0.915	0.664	0.825

Supplementary Figure 1. (A) A heatmap showing a color scale representation of scores for each algorithm (columns) for each variant (rows). For brevity, “CHASM+ Specific” is used for “CHASMplus Cancer Type Specific” and “CHASM+ General” is used for “CHASMplus General”. For all algorithms, a higher score is shown as more red and lower score more yellow. Colored bars on the left side of the plot indicate the Oncogenicity SOP classification (dark red = Oncogenic, orange = Likely Oncogenic, grey = VUS, light blue = Likely Benign, dark blue = Benign, white = score unavailable). The variant rows are sorted by Oncogenicity SOP points with highest points at the top. Variant names on the right describe the gene and amino acid change. (B) A heatmap showing a color scale representation of the Spearman correlations between all possible pairwise comparisons of algorithm scores. For example, scores from MetaLR and FATHMM have a spearman correlation of 0.941. Strong positive correlations are more red, strong negative correlations are more blue, and poor correlations are white. (C) Individual scores are plotted on the y-axis on a scale of 0 to 1 (with higher numbers representing greater predicted effect) for all 70 missense SNVs for all algorithms (x-axis). Variants are color coded according to their Oncogenicity SOP classification as indicated in the legend. (D) An example ROC plot for the VEST4 algorithm. (E) Summary statistics for the performance of all algorithms on the set of 70 missense SNVs including Spearman correlations comparing algorithm score to Oncogenicity SOP points, positive predictive value (PPV) and negative predictive value (NPV) comparing each tool’s recommended interpretation (where available) to the Oncogenicity SOP classification, and the receiver operator characteristic (ROC) area under the curve (AUC) value. Rows are sorted by the ROC AUC value.